Day 2: Routers and Switches

Duration: 1 Hour

Theory Session (30 Minutes)

Topic: Basic Functions and Differences Between Routers and Switches

Learning Objectives

After completing this session, students will be able to:

- Understand the purpose of routers and switches.
- Identify the functions of routers and switches.
- Differentiate between routers and switches.
- Recognize common ports and indicator LEDs.
- Understand how routers and switches work together in a network.

Introduction

Routers and switches are essential networking devices used in homes, offices, schools, and data centers. They help devices communicate with each other and access the internet.

What is a Router?

A router is a networking device that connects different networks and directs data between them.

Functions of a Router

• Connects a local network to the Internet.
• Assigns IP addresses to devices.
• Routes data between networks.
• Provides wired and wireless connectivity.
• Offers basic network security through firewall features.

Common Router Ports

• WAN Port – Connects to the modem or internet source.
• LAN Ports – Connect computers and other devices.
• Power Port – Supplies power to the router.
• Reset Button – Restores factory settings.

Router LEDs

Power LED
• Indicates whether the router is powered on.

Internet/WAN LED
• Shows internet connection status.

Wi-Fi LED
• Indicates wireless network activity.

LAN LED
• Shows activity on connected LAN ports.

What is a Switch?

A switch is a networking device used to connect multiple devices within the same local network.

Functions of a Switch

- Connects computers, printers, and servers.
- Transfers data between devices.
- Reduces unnecessary network traffic.
- Improves network performance.

Common Switch Ports

- Ethernet Ports (RJ45)
- Uplink Port (available on some switches)
- Power Port

Switch LEDs

Power LED
- Indicates power status.

Link LED
- Indicates that a device is connected.

Activity LED
- Blinks when data is being transmitted.

Difference Between Router and Switch

Router

- Connects different networks.
- Provides internet access.

• Assigns IP addresses.
• Operates mainly at Network Layer (Layer 3).

Switch

• Connects devices within the same network.
• Does not provide internet directly.
• Uses MAC addresses for communication.
• Operates mainly at Data Link Layer (Layer 2).

Example Network Setup

Internet

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Modem

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Router

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Switch

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Computers, Printers, and Other Devices

Key Points

• Every network usually requires a router for internet access.
• A switch is used when more wired connections are

needed.
• Routers and switches often work together in business networks.

Practical Session (30 Minutes)

Topic: Identifying Ports and LEDs on a Router and Switch

Objective

Students will identify the physical components of a router and switch and understand their functions.

Requirements

• Router
• Network Switch
• Ethernet Cable
• Computer or Laptop

Activity 1: Router Inspection

Step 1

Observe the router and identify:

• WAN Port
• LAN Ports
• Power Connector
• Reset Button
• Antennas (if available)

Step 2

Power on the router.

Step 3

Observe LED indicators and record their status.

Questions

1. How many LAN ports are available?
2. What is the purpose of the WAN port?
3. Which LED indicates internet connectivity?

Activity 2: Switch Inspection

Step 1

Observe the switch and identify:

• Ethernet Ports
• Power Connector
• Link/Activity LEDs

Step 2

Connect a computer to the switch using an Ethernet cable.

Step 3

Observe the Link LED.

Step 4

Transfer a file or access a network resource and observe the Activity LED blinking.

Questions

1. How many Ethernet ports are available on the switch?
2. Which LED indicates an active network connection?
3. What happens to the Activity LED during data transfer?

Activity 3: Router to Switch Connection

Step 1

Connect one LAN port of the router to any port on the switch using an Ethernet cable.

Step 2

Connect a computer to the switch.

Step 3

Verify that the computer receives network connectivity.

Step 4

Observe the LEDs on both devices.

Expected Result

- Router powers on successfully.
- Switch powers on successfully.
- Connected ports show active Link LEDs.
- Activity LEDs blink during data transfer.
- Students can identify all major ports and indicators.

Conclusion

Routers and switches are fundamental networking devices. Understanding their ports, LEDs, and functions helps technicians install, maintain, and troubleshoot computer networks effectively.